SAQIB ALI



ELECTRICAL ENGINEERING | MOTOR DRIVES | EMBEDDED SYSTEMS

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LinkedIn | GitHub | Portfolio

PROFESSIONAL PROFILE

Electrical Engineering undergraduate at ZHCET, Aligarh Muslim University (expected 2027), with applied experience in five-phase IPMSM control, embedded systems and satellite ground stations. Research intern at IIT Ropar, Ground Station Engineer at Orbitalink and Chairperson of IEEE Student Branch, AMU.

TECHNICAL SKILLS

Control & Drives: MATLAB/Simulink, five-phase IPMSM, FOC, MTPA, flux weakening

Embedded & Power: Arduino, I2C, UART, STM32CubeIDE

RF, Design & Software: GNU Radio, 435 MHz Yagi antennas, MMANA-GAL/NEC, Fusion 360, Altium Designer, KiCad, Python

PROFESSIONAL EXPERIENCE

Research Intern | IIT Ropar, Electrical Engineering Jun–Jul 2026

- Modelled a five-phase IPMSM in MATLAB/Simulink using Clarke–Park transforms; developed dq/xy dual-plane FOC, PI loops, loss-aware MTPA (Newton–Raphson), flux weakening and five-phase SVPWM.

Ground Station Engineer (Part-time) | Orbitalink Pvt. Ltd. Aug 2025–Present

- Lead development and integration of a two-axis azimuth–elevation satellite antenna tracker, spanning hardware, RF design, motion control and firmware.
- Designed a 435 MHz cross-Yagi antenna and Fusion 360 chassis; programmed NEMA23/HSD57 servo drives with trapezoidal ramps, EEPROM persistence and serial control; migrating to STM32F411.

Electrical Design Intern | Orbitalink / AMU Innovation Foundation Jun–Jul 2025

- Implemented adaptive PID pointing control for a high-inertia antenna tracker on Arduino Uno using AS5600 magnetic encoders over I2C.

LEADERSHIP EXPERIENCE

Chairperson | IEEE Student Branch, AMU Aug 2026–Present

- Lead the student branch and its office-bearer team; coordinate technical events, workshops and IEEE member

SELECTED PROJECTS & ACTIVITIES

UHF Satellite Image Transfer & SDR Antenna Testing (2026) — Assisted with testing and validation of a 435 MHz BPSK image-transfer link using GNU Radio, SDR and Python telemetry scripts.

STM32 Test Board (2026) — Designed a PCB with USB-C power, a buck converter and motion sensing in Altium Designer.

437 MHz 9-Element Long-Yagi (2026) — Designed a 50-ohm-feed long-Yagi with a 1:1 balun and construction documentation.

SS AMU SAT Ground Station Team (2024–Present) — Developed an antenna-tracker chassis and two-axis PID logic; support tracking tests and RF equipment diagnostics.

DC-DC Boost Converter Modelling (2025) — Modelled and analysed a DC-DC boost converter in MATLAB/Simulink under varying duty cycles and load conditions.

EDUCATION

B.Tech, Electrical Engineering — Zakir Husain College of Engineering & Technology, Aligarh Muslim University | Sep 2023–Jun 2027

SELECTED CERTIFICATIONS

GE Aerospace Electrical Engineering Job Simulation (2026) | MATLAB Simulation and Control, IEEE STB AMU (2025) | OpenCV Bootcamp (2024)